



In class, we discussed fitting human blood viscosity data using the Sisko Model for a shear-rate-dependent viscosity $\eta(\dot{\gamma})$:

$$\eta - \eta_{\infty} = m \dot{\gamma}^{n-1}$$

where η_{∞} , m , and n are three adjustable constants. It is known that for human blood that $n = 0.5$ is a good approximation.

- (a) Assuming that $n = 0.5$, how can the Sisko model be linearized so that linear least squares can be used to determine the remaining two constants? Show clearly what you would plot as x on the horizontal axis and y on the vertical axis.
- (b) Use the data provided to calculate the remaining two parameters using horizontal offsets. Be sure to give the correct units.

For horizontal offsets: slope $a = \frac{S_{yy} - S_y^2}{S_{xy} - S_x S_y}$ and intercept $b = \frac{S_{xy} S_y - S_{yy} S_x}{S_{xy} - S_x S_y}$.

$n \neq \eta$

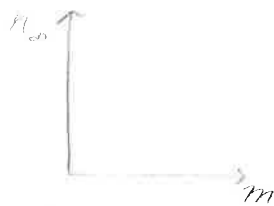
(a) $0.5 - n_{\infty} = m \dot{\gamma}^{0.5-1}$

$$0.5 - n_{\infty} = m \dot{\gamma}^{-0.5}$$

$$n_{\infty} = \frac{m \dot{\gamma}^{-0.5}}{-0.5}$$

$$n_{\infty} = \frac{-m}{(0.5) \gamma_{0.5}}$$

$$0.5 - n_{\infty} = m \dot{\gamma}^{0.5-1}$$
$$0.5 - n_{\infty} = m \dot{\gamma}^{-0.5}$$



$$S_x = 171.924 (s^{-1})$$

$$S_y = 16.168 (mPa \cdot s) \quad S_{xy} = 3859.8606 (s^{-1} mPa \cdot s)$$

Shear Rate $\dot{\gamma} (s^{-1})$	Viscosity $\eta (mPa \cdot s)$
0.51	39.15
1.34	22.18
7.97	9.51
90.73	5.77
759.07	4.23

$$S_{xx} = 116896.9549$$

$x_i y_i$	x_i^2	y_i^2
19.9665		
29.7212		
75.7947		
523.5121		
3210.8661		

$$S_{yy} = 2166.3008 (mPa \cdot s)^2$$

(b)

$$a = \frac{S_{yy} - S_y^2}{S_{xy} - S_x S_y}$$

$$a = \frac{2166.3008 - (16.168)^2}{3859.8606 - 2779.667232}$$

$$a = \frac{1904.896576}{1080.193368}$$

$$a = 1.76$$

$$(mPa \cdot s)^2 - (mPa \cdot s)^2$$

$$(s^{-1} mPa \cdot s) - (s^{-1} mPa \cdot s)$$

[=]

$$b = \frac{S_{xy} S_y - S_{yy} S_x}{S_{xy} - S_x S_y}$$

$$b = \frac{3859.8606 (16.168) - 2166.3008 (171.924)}{3859.8606 - 2779.667232}$$

$$b = \frac{-310032.8726}{1080.193368}$$

$$b = -287.016$$

$$((s^{-1} mPa \cdot s)) (mPa \cdot s) - (mPa \cdot s)^2 (s^{-1})$$

[=]

$$(s^{-1} mPa \cdot s) - s^{-1} (mPa \cdot s)$$

